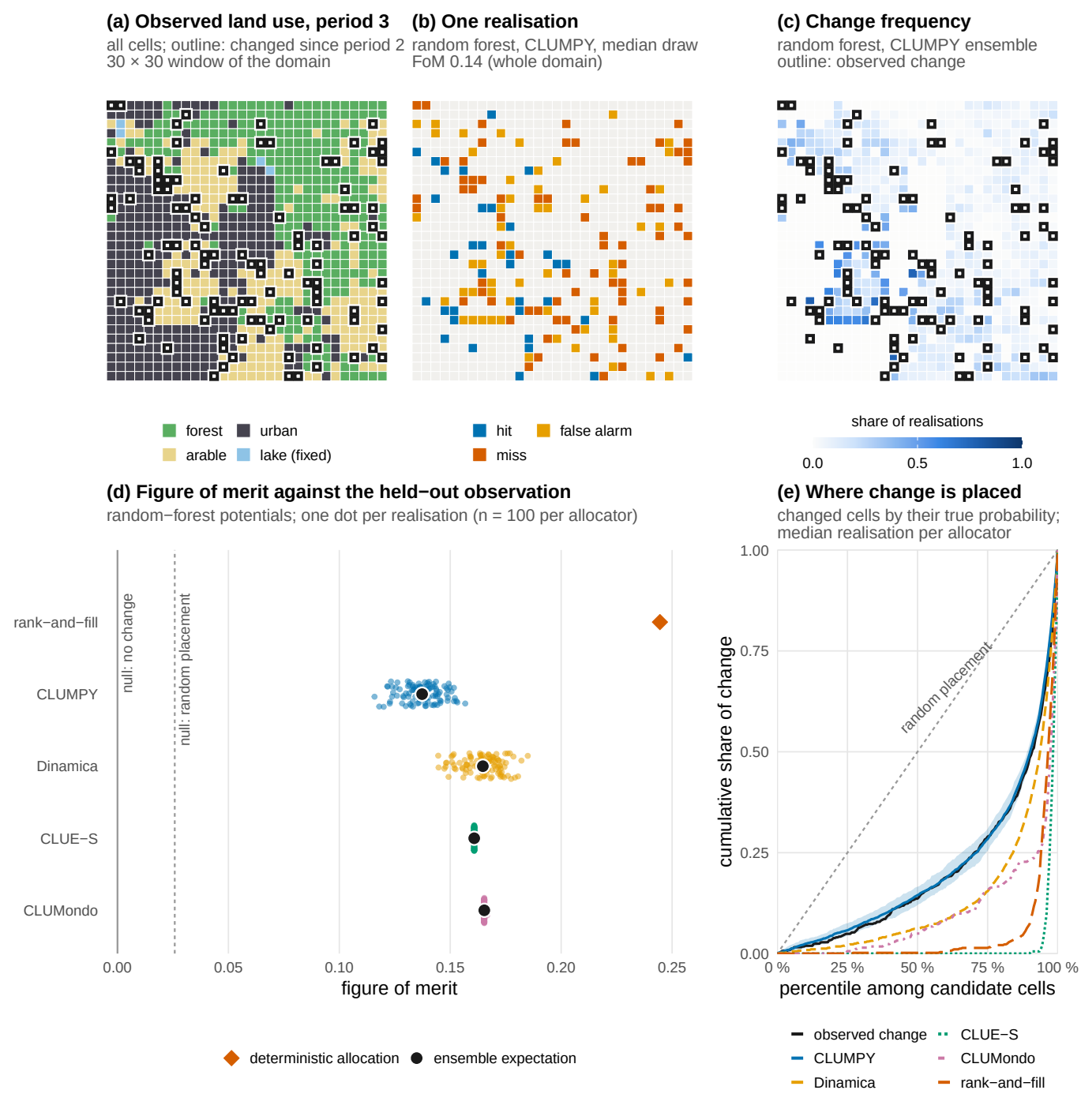
(a) Observed land use, period 3
all cells; outline: changed since period 2
30 × 30 window of the domain
(b) One realisation
random forest, CLUMPY, median draw
FoM 0.14 (whole domain)
(c) Change frequency
random forest, CLUMPY ensemble
outline: observed change
forest
urban
arable
lake (fixed)
hit
false alarm
miss
share of realisations
0.0
0.5
1.0
(d) Figure of merit against the held−out observation
random−forest potentials; one dot per realisation (n = 100 per allocator)
rank−and−fill
CLUMPY
Dinamica
CLUE−S
CLUMondo
null: no change
null: random placement
0.00
0.05
0.10
0.15
0.20
0.25
figure of merit
deterministic allocation
ensemble expectation
(e) Where change is placed
changed cells by their true probability;
median realisation per allocator
cumulative share of change
1.00
0.75
0.50
0.25
0.00
random placement
0 %
25 %
50 %
75 %
100 %
percentile among candidate cells
observed change
CLUMPY
Dinamica
CLUE−S
CLUMondo
rank−and−fill